



Career Development Training Center

2024-25

Python with Data Science



Python with Data Science - Complete 8-Month Syllabus

Duration: 6 Months Training+2 Month Internship

Mode: Online/Offline

Prerequisites: Basic computer knowledge

Module 1: Python Programming Basics (Weeks 1–4)

- **Introduction to Python**

- ✓ Python features, installation, and setup
- ✓ Python IDEs (Jupyter, PyCharm, VS Code)
- ✓ Writing and running Python scripts

- **Core Python Concepts**

- ✓ Environment Setup
- ✓ Basic Syntax
- ✓ Variable Types
- ✓ Basic Operators
- ✓ Decision Making
- ✓ Loops
- ✓ Numbers
- ✓ Strings
- ✓ Lists
- ✓ Tuples



- ✓ **Dictionary**
- ✓ **Date & Time**
- ✓ **Functions**
- ✓ **Modules**
- ✓ **Files I/O**
- ✓ **Exceptions**

- **Working with Data Structures**

- ✓ **Lists, tuples, sets, and dictionaries**
- ✓ **List comprehensions and dictionary comprehensions**
- ✓ **String manipulation**

- **Advance Python Programming**

- ✓ **POP+OPPS Concept**
- ✓ **Reg Expressions**
- ✓ **Database Access**
- ✓ **Sending Email**
- ✓ **Multithreading**
- ✓ **XML Processing**
- ✓ **GUI Programming**
- ✓ **MYSQL Database Server**

Module 2: Data Analysis with Python (Weeks 5–8)

Numpy for Numerical Computing

- ✓ Introduction to NumPy
- ✓ Basics of NumPy Arrays
- ✓ Array Indexing and Slicing
- ✓ Array Operations
- ✓ Array Manipulation
- ✓ Statistical and
Mathematical Operations
- ✓ Working with Random Data
- ✓ Linear Algebra with NumPy
- ✓ Handling Missing Data
- ✓ Working with Files
- ✓ Advanced Topics
- ✓ Applications of NumPy
- ✓ Practice Exercises and Projects



INFOTECH

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Pandas for Data Manipulation

- **Introduction to Pandas**
- **Series and DataFrames**
- **Data Input/Output (CSV, Excel, SQL, JSON)**
- **Data Selection and Indexing**
- **Data Manipulation and Cleaning**
- **Handling Missing Data**
- **Data Filtering and Conditional Selection**
- **Data Aggregation and Grouping**
- **Merging, Joining, and Concatenation**
- **Pivot Tables and Crosstabs**
- **Working with Time-Series Data**
- **Sorting and Ranking Data**
- **Iterating through DataFrames**
- **Working with Text Data**
- **Visualization with Pandas**
- **Integration with Matplotlib and Seaborn**
- **Exporting Data to Files**
- **Performance Optimization in Pandas**
- **Advanced Features: MultiIndexing and Reshaping Data**

Data Visualization

Matplotlib:

Plot types, customization, and subplots

Seaborn:

Advanced visualizations (heatmaps, pair plots)

Module 3: Statistics and Probability for Data Science (Weeks 9–12)

- **Basic Statistics**

- ✓ Measures of central tendency (mean, median, mode)
- ✓ Measures of variability (variance, standard deviation, IQR)
- ✓ Probability distributions: Normal, binomial, Poisson

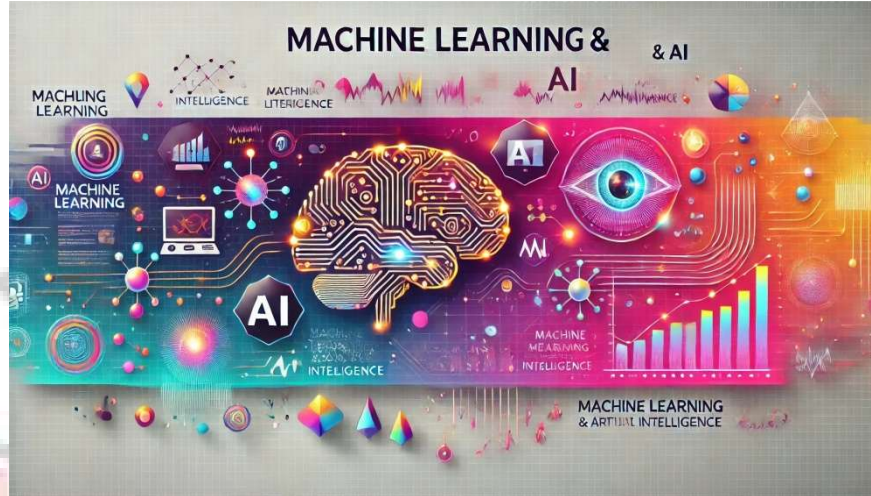
- **Hypothesis Testing**

- ✓ Null and alternative hypotheses
 - ✓ Z-test, t-test, chi-square test
 - ✓ ANOVA and correlation analysis
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Module 4: Machine Learning Basics (Weeks 13–16)

Introduction to Machine Learning

- ✓ **Types: Supervised, unsupervised, reinforcement learning**
- ✓ **Machine learning workflow**
- **Supervised Learning**
 - ✓ **Regression (Linear, Logistic)**
 - ✓ **Classification (KNN, Decision Trees, SVM)**
- **Unsupervised Learning**
 - ✓ **Clustering (K-Means, Hierarchical Clustering)**
 - ✓ **Dimensionality reduction (PCA, t-SNE)**



Module 5: Advanced Python for Data Science (Weeks 17–20)

- **Web Scraping with Python**
 - ✓ **BeautifulSoup, Scrapy**
 - ✓ **Automating data collection with APIs (e.g., Twitter, Google Maps)**

Time Series Analysis

- ✓ **Components of time series**
- ✓ **ARIMA, SARIMA models**

- **Introduction to Deep Learning**

- ✓ **Basics of neural networks**
- ✓ **TensorFlow and Keras overview**

Module 6: Real-World Applications and Capstone Project (Weeks 21–24)

- **Real-World Applications**

- ✓ **Data pipelines with Python**
- ✓ **Deploying models using Flask or FastAPI**
- ✓ **Case studies: Business analytics, customer segmentation**

- **Capstone Project**

- ✓ **End-to-end project on a real-world dataset**
 - ✓ **Problem statement, EDA, model building, and deployment**
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Additional Resources

- Mock interviews and assessments
- Regular assignments and Programs
- Access to datasets and notebooks
- Certificate of Completion

Job Details:

As of 2024–2025, the demand for data scientists in India continues to grow, offering promising opportunities for beginners in the field. Here's an overview of current job prospects and expected salary packages for entry-level data scientists:

Job Opportunities for Beginners:

Entry-level positions in data science are available across various industries, including technology, finance, healthcare, and e-commerce. Common job titles for beginners include:

- **Data Analyst**
- **Junior Data Scientist**
- **Business Intelligence Analyst**
- **Machine Learning Intern**

Salary Packages:

Salaries for entry-level data science roles in India vary based on factors such as location, industry, and individual skill sets. As of 2024–2025, the estimated salary ranges are:

- **Average Base Salary:** ₹6,00,000 per annum
- **Salary Range:** ₹4,00,000 – ₹10,00,000 per annum